

Optimal Modeling of Surgical Planning

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1 Introduction

This document presents a linear programming modeling for the surgery planning problem. The objective is to minimize the fixed costs of using operating rooms and the variable costs related to overtime, while respecting a set of operational constraints.

Project Context: This report is a Version 2 (V2) of the original project conducted in December 2025 by the four authors (Toscane Fantoni, Hannah Guerrault, Colin Borel, Gabin Lemaître). It has been significantly enriched and expanded in this version by designing and implementing a novel **Hybrid Stochastic-Robust model (Light Robustness)** to overcome the theoretical limitations identified in the classic approaches.

2 Sets and Parameters

- $I = \{1, \dots, n\}$: set of patients.
- J : set of surgical specialties.
- $K = \{1, \dots, 7\}$: set of days of the week (1-Monday, 6-Saturday, 7-Sunday).
- $s_i \in J$: specialty of patient $i \in I$.
- $t_{i,fixe}$: minimum deterministic duration of the operation for patient i .
- $t_{i,sup}$: maximum possible additional duration for patient i .
- C_j^{base} : fixed cost for specialty j during the week.
- $C_j'^{base}$: additional hourly cost for specialty j during the week.
- $H = 8$: number of standard working hours per day.
- $H_{max_sup} = 2$: maximum allowed overtime hours.

Calculation of daily costs

The effective costs C_{jk} and C'_{jk} depend on day k :

- Weekdays ($k \in \{1, \dots, 5\}$): $C_{jk} = C_j^{base}$, $C'_{jk} = C_j'^{base}$.
- Saturday ($k = 6$): $C_{j6} = 1.25C_j^{base}$, $C'_{j6} = 1.25C_j'^{base}$.
- Sunday ($k = 7$): $C_{j7} = 1.8C_j^{base}$, $C'_{j7} = 1.8C_j'^{base}$.

3 Decision Variables

- $x_{ik} = 1$ if patient i is operated on day k , 0 otherwise.
- $y_{jk} = 1$ if the room is allocated to specialty j on day k , 0 otherwise.
- $\tau_{jk} \geq 0$: overtime hours of specialty j on day k .

4 Deterministic Modeling

4.1 Objective Function

$$\min Z = \sum_{j \in J} \sum_{k \in K} C_{jk} y_{jk} + \sum_{j \in J} \sum_{k \in K} C'_{jk} \tau_{jk}$$

4.2 Constraints

1. Each patient is operated on exactly once:

$$\sum_{k \in K} x_{ik} = 1 \quad \forall i \in I$$

2. Only one specialty per day:

$$\sum_{j \in J} y_{jk} \leq 1 \quad \forall k \in K$$

3. Specialty consistency:

$$x_{ik} \leq y_{s_i, k} \quad \forall i \in I, k \in K$$

4. Definition of overtime:

$$\sum_{i \in I | s_i = j} t_{i, fixe} x_{ik} - H y_{jk} \leq \tau_{jk} \quad \forall j \in J, k \in K$$

5. Maximum bound:

$$0 \leq \tau_{jk} \leq H_{max-sup} y_{jk} \quad \forall j, k$$

6. Nature of variables: $x_{ik}, y_{jk} \in \{0, 1\}$.

5 Stochastic Modeling

5.1 Introduction of uncertainty

In practice, actual surgical intervention durations are difficult to predict, especially due to unforeseen operative events. In order to represent this variability, we introduce a scenario-based stochastic modeling. For each patient i , the operative time is decomposed into two parameters:

- $t_{i, fixe}$: the minimal and deterministic operative time needed.
- $t_{i, sup}$: the maximum additional time that patient i 's intervention may require.

In each scenario $\omega \in \Omega$, the actual duration of the operation is modeled by the random variable:

$$t_i^\omega = t_{i, fixe} + \alpha_i^\omega t_{i, sup}$$

where $\alpha_i^\omega \sim \mathcal{U}[0, 1]$.

5.2 Two-stage decisions and SAA Formulation

The stochastic model remains a two-stage program. Using the Sample Average Approximation (SAA) with a finite set of scenarios $\Omega = \{\omega_1, \dots, \omega_S\}$ with probabilities $p_s = 1/S$, the model becomes a single large linear program:

$$\min Z_{SAA} = \sum_{j \in J} \sum_{k \in K} C_{jk} y_{jk} + \sum_{s=1}^S p_s \left(\sum_{j \in J} \sum_{k \in K} C'_{jk} \tau_{jk}^s \right)$$

subject to constraints:

$$\begin{aligned} \sum_{i: s_i = j} (t_{i, fixe} + \alpha_i^s t_{i, sup}) x_{ik} - H y_{jk} &\leq \tau_{jk}^s, \quad \forall j, k, s, \\ 0 &\leq \tau_{jk}^s \leq H_{max-sup} y_{jk}, \quad \forall j, k, s. \end{aligned}$$

6 Robust Modeling

6.1 Motivation and Limits of the Stochastic Approach

The stochastic approach minimizes an expected cost. Although efficient on average, it presents a major risk regarding operational feasibility. A stochastically optimized schedule can lead, in "tail-risk" scenarios, to total durations exceeding the maximum allowable physical limit (10 hours), resulting in canceled surgeries.

Robust Optimization (Bertsimas & Sim) aims to immunize the schedule against these uncertainties by protecting against a defined **uncertainty budget** Γ .

6.2 Final Robust Model

Using strong duality, the maximization of the worst-case delay is linearized using dual variables q_{jk} and r_{ik} . The robust capacity constraint becomes:

$$\sum_{i \in I | s_i = j} t_{i,fixe} x_{ik} + \Gamma q_{jk} + \sum_{i \in I | s_i = j} r_{ik} - H y_{jk} \leq \tau_{jk} \quad \forall j \in J, k \in K$$

$$q_{jk} + r_{ik} \geq t_{i,sup} x_{ik} \quad \forall i \in I, j = s_i, k \in K$$

6.3 Analysis of the uncertainty budget and the "Dense-Packing" effect

Simulation of this model over 100 scenarios revealed a counter-intuitive flaw in the classic robust approach. While $\Gamma = 2$ provided excellent reliability, **reliability paradoxically decreased for $\Gamma \geq 3$** .

This is due to a mathematical loophole we identified as the **"Dense-Packing" effect**. When pushed to guarantee protection against many simultaneous maximum delays, the solver avoids the exorbitant fixed costs of opening weekend rooms by isolating high-risk patients and densely packing many "low-risk" patients into the remaining weekday slots.

Gamma_Model	Day_Index	Day_Name	Specialty	Patient_ID	Base_Duration_H	Overtime_Retained_I	Max_Potential_Overt
3	1	Monday	ORTHO	P12	0,36	0,46	0,46
3	1	Monday	ORTHO	P32	0,56	0,47	0,47
3	1	Monday	ORTHO	P35	1,2	0,5	0,5
3	1	Monday	ORTHO	P36	1,13	0	0,37
3	1	Monday	ORTHO	P42	1,07	0	0,39
3	1	Monday	ORTHO	P43	1,02	0	0,42
3	1	Monday	ORTHO	P47	0,58	0	0,38
3	1	Monday	ORTHO	P50	0,75	0	0,36
3	2	Tuesday	VASCULAR	P9	1,13	0,39	0,39
3	2	Tuesday	VASCULAR	P24	2,62	0,49	0,49
3	2	Tuesday	VASCULAR	P30	1,05	0,31	0,31
3	2	Tuesday	VASCULAR	P44	0,75	0	0,28
3	2	Tuesday	VASCULAR	P45	1,88	0	0,28
3	3	Wednesday	ORTHO	P0	0,46	0	0,27
3	3	Wednesday	ORTHO	P2	0,52	0	0,31
3	3	Wednesday	ORTHO	P11	0,74	0	0,31
3	3	Wednesday	ORTHO	P13	1	0,35	0,35
3	3	Wednesday	ORTHO	P20	0,63	0	0,28
3	3	Wednesday	ORTHO	P25	0,89	0,32	0,32
3	3	Wednesday	ORTHO	P26	1,61	0	0,23
3	3	Wednesday	ORTHO	P28	0,82	0	0,29
3	3	Wednesday	ORTHO	P40	0,56	0,32	0,32

Figure 1: Uneven distribution of overtime across days leading to oscillating results.

As illustrated in Figure 1, the solver's strategy leads to highly unbalanced schedules. For instance, the generated overtime on Day 2 is significantly shorter than on other days. Because the objective function only minimizes the *total* cost, the solver opportunistically shifts burdens across days, creating extreme variances. Since the robust model assumes patients not covered by Γ experience *zero* delay, it overestimates the safety of the densely packed days. In reality, the accumulation of minor delays on the heavily loaded days quickly breaches the 10-hour limit, causing the schedule to crash and leading to the oscillating ("yo-yo") reliability results observed.

7 Hybrid Modeling (Light Robustness) - V2 Addition

To prevent the solver from mathematically exploiting the uncertainty budget and dense-packing patients, we developed a **Hybrid Stochastic-Robust model**.

7.1 Mathematical Adjustment

Instead of assuming that patients outside the Γ budget will have zero delay, we shift the baseline duration of every patient to their *expected* duration. Assuming delays follow a uniform distribution $\mathcal{U}(0, \alpha_{max})$, the adjusted baseline is:

$$t_{i,base_adj} = t_{i,fixe} + \left(\frac{\alpha_{max}}{2} \cdot t_{i,sup} \right)$$

The uncertainty budget Γ is then strictly used to protect against the *variance* (the extreme residual tail-end of the distribution):

$$t_{i,sup_adj} = \left(1 - \frac{\alpha_{max}}{2} \right) \cdot t_{i,sup}$$

7.2 Results

By replacing $t_{i,fixe}$ and $t_{i,sup}$ with their adjusted counterparts in the robust constraints, the model is forced to acknowledge the real statistical weight of every patient. This definitively solves the "dense-packing" issue. The reliability no longer collapses at high Γ values, and the resulting schedules are operationally balanced and highly resilient.

8 Conclusion

The evolution of our modeling from a deterministic baseline to a hybrid formulation provides a comprehensive framework for surgical scheduling under uncertainty:

- **The Deterministic model** is highly cost-efficient but operationally unviable, failing in nearly 60% of simulated real-world scenarios due to over-optimism.
- **The Stochastic model (SAA)** offers a good compromise for minimizing average financial costs but remains vulnerable to severe, simultaneous delays (tail risks) that can force the hospital to cancel procedures.
- **The Classic Robust model** provides strong guarantees but becomes brittle and overly conservative at higher protection budgets due to the dense-packing exploit.
- **The Hybrid model** stands out as the ultimate solution. By combining the statistical realism of the stochastic expectation with the hard bounds of robust optimization, it creates a Pareto-optimal schedule.

Strategic Recommendation: We conclude that the **Hybrid model configured with an uncertainty budget of $\Gamma = 2$** is the optimal strategic choice for the hospital. It guarantees near 100% reliability across standard and extreme stress levels ($\alpha = 1.0$ to 1.5) while minimizing the structural cost of unnecessary weekend room openings.

8.1 Perspectives for improvement

Several avenues for further improvement can be considered:

1. **Non-Integer Uncertainty Budgets:** Refining the robust protection by allowing fractional values for Γ (e.g., $\Gamma = 1.5$) could offer an even finer trade-off between cost and reliability.
2. **Advanced Probability Distributions:** While the uniform distribution $\mathcal{U}[0, \alpha]$ is a solid baseline, surgical delays in reality often follow log-normal or Weibull distributions. Adapting the hybrid baseline calculation to these specific distributions would further enhance real-world accuracy.
3. **Specialty-Specific Severity:** Applying different α_{max} parameters per surgical specialty (e.g., higher variance for Vascular surgery compared to ENT) would allow the model to allocate safety buffers more intelligently.